Thermal mechanics of floppy networks of rigid bars

We consider floppy networks in $d=3$ dimensions with $z<z_c=2d$, of unit masses connected by perfectly rigid, massless bars. We require

$$\mathcal{S}|v\rangle \equiv \sum_\alpha \frac{\partial r_\alpha}{\partial \boldsymbol{x}_k}\cdot \boldsymbol{v}_k = 0\,, \tag{1}$$

where $\alpha=(i,j)$ represents a bar. We also require that the time-derivative vanishes, namely

$$\frac{d}{dt}\Big(\mathcal{S}|v\rangle\Big) = \dot{\mathcal{S}}|v\rangle + \mathcal{S}|a\rangle = \boldsymbol{\mathcal{K}}:\boldsymbol{vv} + \mathcal{S}|a\rangle = 0\,, \tag{2}$$

where $\boldsymbol{a}$ denotes the nodes' accelerations, and

$$\boldsymbol{\mathcal{K}} \equiv \frac{\partial^2 r_\alpha}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell} = (\delta_{jk}-\delta_{ik})(\delta_{j\ell}-\delta_{i\ell})\left(\frac{\boldsymbol{\mathcal{I}}}{r_\alpha} - \frac{\boldsymbol{x}_\alpha \boldsymbol{x}_\alpha}{r_\alpha^3}\right). \tag{3}$$

The acceleration must result from forces $|f\rangle$ in the bars, and so Eq. (2) should read

$$|\boldsymbol{\mathcal{K}}:\boldsymbol{vv}\rangle + \mathcal{S}\mathcal{S}^T|f\rangle = 0\,. \tag{4}$$

This equation can be inverted in favor of the forces, as

$$|f\rangle = -\big(\mathcal{S}\mathcal{S}^T\big)^{-1}|\boldsymbol{\mathcal{K}}:\boldsymbol{vv}\rangle\,, \tag{5}$$

then the accelerations are given by

$$|a\rangle = \mathcal{S}^T|f\rangle = -\mathcal{S}^T\big(\mathcal{S}\mathcal{S}^T\big)^{-1}|\boldsymbol{\mathcal{K}}:\boldsymbol{vv}\rangle\,. \tag{6}$$

1 Setup and Notation

We consider a network of N unit-mass nodes in $\bar{d}$ dimensions, connected by N_c perfectly rigid, massless bars. Node positions are collected into a $N\bar{d}$-dimensional vector $|x\rangle$, with velocities $|v\rangle$ and accelerations $|a\rangle$. Bar $\alpha=(i,j)$ has equilibrium length ℓ_α, current length $r_\alpha=|\boldsymbol{x}_j-\boldsymbol{x}_i|$, and unit vector $\hat{n}_\alpha=(\boldsymbol{x}_j-\boldsymbol{x}_i)/r_\alpha$.

The rigidity matrix $\mathcal{S}$ maps node velocities to rates of change of bar lengths,

$$\mathcal{S}_{\alpha k} = \frac{\partial r_\alpha}{\partial x_k} = (\delta_{jk}-\delta_{ik})\,\hat{n}_\alpha\,, \tag{7}$$

so that $|\dot{r}\rangle=\mathcal{S}|v\rangle$. The constraint that bar lengths are preserved requires

$$\mathcal{S}|v\rangle = 0\,. \tag{8}$$

The curvature tensor of bar α is

$$K_\alpha^{k\ell} = \frac{\partial^2 r_\alpha}{\partial x_k\,\partial x_\ell} = (\delta_{jk}-\delta_{ik})(\delta_{j\ell}-\delta_{i\ell})\left(\frac{I}{r_\alpha} - \frac{\hat{n}_\alpha\hat{n}_\alpha}{r_\alpha}\right). \tag{9}$$

We define the symmetric positive semi-definite matrix

$$\mathcal{N} \equiv \mathcal{S}\mathcal{S}^T\,, \tag{10}$$

and denote by $|K{:}vv\rangle$ the contact-space vector with components $(K{:}vv)_\alpha = K_\alpha^{k\ell}v_k v_\ell$.

2 Equation of Motion

The Hamiltonian is purely kinetic,

$$H = \tfrac{1}{2}\langle v|v\rangle\big|_{\mathcal{S}|v\rangle=0}\,, \tag{11}$$

since the bars carry no potential energy. The acceleration is determined by requiring that the time derivative of the constraint (8) vanishes, giving

$$|a\rangle = -\mathcal{S}^T\mathcal{N}^{-1}|K{:}vv\rangle\,. \tag{12}$$

This is the projection of the centripetal geometric force onto the constraint manifold $\mathcal{M} = \{x : r_\alpha(x) = \ell_\alpha\ \forall\,\alpha\}$. There is no potential energy contribution; the right-hand side arises entirely from the curvature of $\mathcal{M}$.

3 Velocity-Verlet (RATTLE) Integration

The constraints must be maintained on both positions and velocities at each timestep. The standard approach is the RATTLE algorithm [**andersen1983rattle**], which is symplectic and time-reversible, ensuring long-time energy conservation without secular drift. The integration step from time t^n to $t^{n+1} = t^n + \delta t$ proceeds as follows.

[H] RATTLE step for rigid bar NVE dynamics [1] **Half-kick:** $|v^{n+1/2}\rangle \leftarrow |v^n\rangle + \frac{\delta t}{2}|a^n\rangle$ **Drift:** $|x^{n+1}\rangle \leftarrow |x^n\rangle + \delta t\,|v^{n+1/2}\rangle$ **Rebuild geometry:** recompute $\mathcal{S}^{n+1}$, $\mathcal{N}^{n+1}$, and $|K^{n+1}{:}v^{n+1/2}v^{n+1/2}\rangle$ **Acceleration:** $|a^{n+1}\rangle \leftarrow -(\mathcal{S}^{n+1})^T(\mathcal{N}^{n+1})^{-1}|K^{n+1}{:}v^{n+1/2}v^{n+1/2}\rangle$ **Half-kick:** $|v^{n+1}\rangle \leftarrow |v^{n+1/2}\rangle + \frac{\delta t}{2}|a^{n+1}\rangle$ **Velocity projection:** $|v^{n+1}\rangle \leftarrow |v^{n+1}\rangle - (\mathcal{S}^{n+1})^T(\mathcal{N}^{n+1})^{-1}\mathcal{S}^{n+1}|v^{n+1}\rangle$

Step 6 removes any velocity component that would violate the constraint $\mathcal{S}|v\rangle = 0$, which is generated at order δt^2 by the drift step. Steps 4 and 6 use $|v^{n+1/2}\rangle$ (mid-step velocities at the updated geometry) to evaluate the $K{:}vv$ term; this is the correct choice since the positions are already at x^{n+1} at that point.

The solve of $\mathcal{N}|f\rangle = |b\rangle$ required in steps 4 and 6 should be performed using a sparse Cholesky or conjugate gradient solver, as $\mathcal{N}$ is sparse but $N\bar{d} \times N\bar{d}$.

4 Position Correction Step

Integration errors accumulate in the bar lengths at rate $O(\delta t^2)$ per step, leading to slow drift of node positions off $\mathcal{M}$. Following Lerner, Düring and Wyart [**lerner2013driven**], we periodically apply the following correction, which eliminates the accumulated bar-length errors via the minimum-norm displacement.

Define the vector of bar-length errors in contact space,

$$(\delta r)_\alpha = r_\alpha(x) - \ell_\alpha\,, \tag{13}$$

and compute the correction displacement

$$|\delta x\rangle = -\mathcal{S}^T\mathcal{N}^{-1}|\delta r\rangle\,. \tag{14}$$

This is the minimum-ℓ^2-norm displacement field in node space that exactly cancels the first-order bar-length errors, as can be verified by acting with $\mathcal{S}$:

$$\mathcal{S}|\delta x\rangle = -\mathcal{S}\mathcal{S}^T\mathcal{N}^{-1}|\delta r\rangle = -|\delta r\rangle\,. \tag{15}$$

After updating $|x\rangle \leftarrow |x\rangle + |\delta x\rangle$, the rigidity matrix changes, so the velocities must be re-projected onto the updated constraint manifold, and the energy must be restored.

[H] Periodic position correction (applied every M steps, or when $\max_\alpha |(\delta r)_\alpha|/\ell_\alpha > \varepsilon_{\rm tol}$) [1] Record kinetic energy $E_0 = \frac{1}{2}\langle v|v\rangle$ Compute bar-length errors: $(\delta r)_\alpha = r_\alpha(x) - \ell_\alpha$ for all α Compute correction: $|\delta x\rangle = -\mathcal{S}^T\mathcal{N}^{-1}|\delta r\rangle$ Update positions: $|x\rangle \leftarrow |x\rangle + |\delta x\rangle$ Rebuild geometry: recompute $\mathcal{S}$, $\mathcal{N}$ Re-project velocities: $|v\rangle \leftarrow |v\rangle - \mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|v\rangle$ Rescale velocities to restore energy: $|v\rangle \leftarrow |v\rangle\sqrt{E_0\,/\,\frac{1}{2}\langle v|v\rangle}$

Remarks on step 7. Since the Hamiltonian is purely kinetic, the total energy is $H = \frac{1}{2}\langle v|v\rangle$, and a global rescaling

$$|v\rangle \leftarrow \lambda|v\rangle, \qquad \lambda = \sqrt{\frac{E_0}{\langle v|v\rangle/2}} \tag{16}$$

restores H exactly with no approximation. Crucially, because the rescaling is a uniform scalar multiple, it commutes with the constraint projection â it does not reintroduce any component along the bars â so no further projection is required after the rescale. The velocity projection in step 6 reduces $\langle v|v\rangle$, so $\lambda \geq 1$ always; a large λ signals that the correction step is doing significant work and that either M or δt should be reduced.

Trigger condition. Rather than applying the correction on a fixed schedule, it is cleaner to trigger it whenever the maximum relative bar-length error exceeds a threshold,

$$\max_{\alpha} \frac{|r_\alpha(x) - \ell_\alpha|}{\ell_\alpha} > \varepsilon_{\text{tol}} \,, \tag{17}$$

since near floppy modes the drift rate varies strongly with configuration. A value $\varepsilon_{\text{tol}} \sim 10^{-8}$ is typical.

5 Diagnostics

Three quantities should be monitored throughout the simulation:

1. **Energy conservation:** $H = \frac{1}{2}\langle v|v\rangle$ should be constant between correction steps. Drift indicates δt is too large.
2. **Bar-length violations:** $\max_\alpha |r_\alpha - \ell_\alpha|/\ell_\alpha$ should remain at or below ε_{tol} immediately after each correction, and drift at rate $O(\delta t^2)$ between corrections.
3. **Velocity constraint:** $\max_\alpha |(\mathcal{S}|v\rangle)_\alpha|$ should remain near machine epsilon after each RATTLE velocity projection. Growth here indicates a bug in the projection step.

6 Summary of Linear Solves

Each timestep and correction step requires solving systems of the form $\mathcal{N}|f\rangle = |b\rangle$ where $\mathcal{N} = \mathcal{S}\mathcal{S}^T$ is $N_c \times N_c$, sparse, symmetric and positive definite (provided the network has no states of self-stress). The velocity projection additionally requires computing $\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|v\rangle$, which can be done in two solves. For a floppy network well below z_c, $\mathcal{N}$ is well-conditioned and a sparse Cholesky factorization is efficient. Near z_c, $\mathcal{N}$ develops small eigenvalues corresponding to soft modes, and a conjugate gradient solver with an appropriate preconditioner is preferable.

References

[1] E. Lerner, G. Düring, and M. Wyart, *Simulations of driven overdamped frictionless hard spheres*, arXiv:1111.7225 (2013).

[2] H. C. Andersen, *RATTLE: A "velocity" version of the SHAKE algorithm for molecular dynamics calculations*, J. Comput. Phys. **52**, 24 (1983).